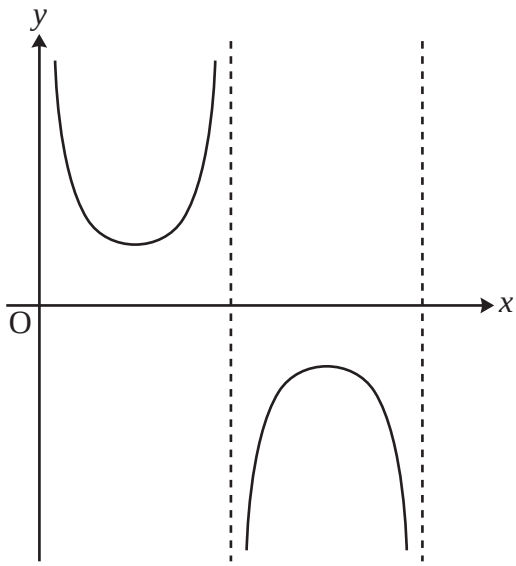
6 **(a)** The diagram shows part of the graph of $y = \text{cosec}\, x$, where x is in radians.

State the equations of the **three** vertical asymptotes that can be seen. **[1]**



The tangent to the graph at the point P with x-coordinate $\frac{\pi}{3}$ meets the x-axis at Q.

(b) Show that the x-coordinate of Q is $\frac{\pi}{3} + \sqrt{3}$. (You may use without proof the result that the derivative of $\text{cosec}\, x$ is $-\text{cosec}\, x \cot x$.) **[6]**